Learning to See the Terrain: Semantic Segmentation for Off-Road Navigation

BY TEAM -MAVERICKS
team Id- 7c928ef3

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Offroad Autonomy Segmentation

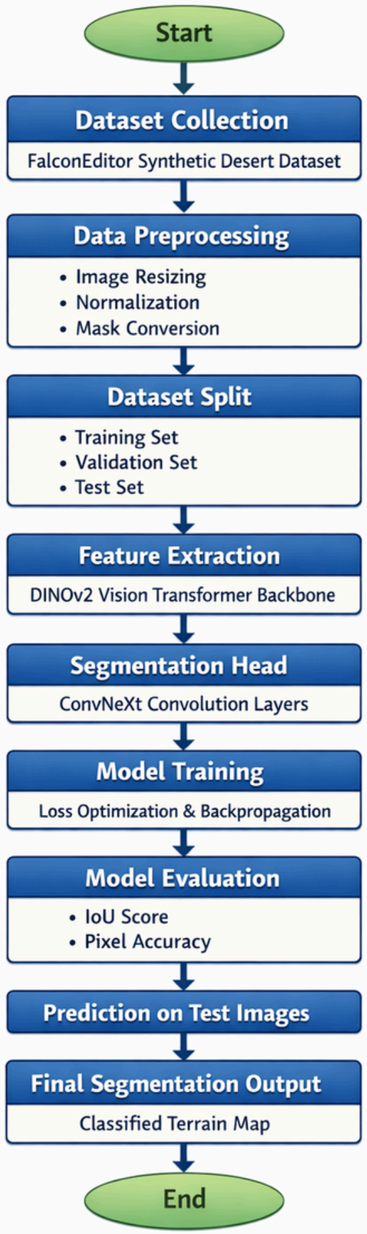
- Organized by Duality AI to explore advanced AI training techniques.
- To develop a semantic segmentation model for desert environments.
- Models are trained using annotated synthetic images generated from the Falcon digital twin simulation platform.
- The trained model is evaluated on a new unseen desert environment to test generalization.
- It focuses on improving accuracy, robustness, and real-world applicability of AI models.

Problem Statement

- Autonomous off-road vehicles require accurate understanding of complex terrain to navigate safely. However, identifying different environmental elements such as trees, bushes, grass, rocks, and sky in desert environments is challenging due to similar textures, varying lighting conditions, and limited labeled real-world data.
- The objective of this project is to develop a robust semantic segmentation model that can accurately classify each pixel in desert environment images into predefined terrain classes using synthetic data generated from Duality AI's Falcon digital twin platform. The trained model should generalize well to unseen desert environments and achieve high segmentation accuracy for reliable off-road perception.

Methodology

1. **Dataset**
   - Generated using FalconEditor (Duality AI digital twin platform).
   - Contains train, validation, and test datasets.
   - Each image has a segmentation mask with pixel-level labels.
2. **Data Preprocessing**
   - Resizing: All images resized to a fixed size.
   - Normalization: Applied ImageNet mean and standard deviation.
   - Mask Conversion: Converted color masks to numerical class IDs.
3. **Model Architecture**
   - Backbone: DINOv2 Vision Transformer for feature extraction.
   - Segmentation Head: ConvNeXt-based layers for pixel-wise predictions.



Model Training

Model Training

- Framework: PyTorch
- Batch Size: 4
- Learning Rate: 0.0001
- Epochs: 25
- Loss Function: Cross Entropy

Training Workflow

- Load training images and segmentation masks
- Extract features using DINOv2
- Train segmentation head
- Compute training loss
- Validate model performance

Evaluation Metrics

- IoU: Measures overlap between prediction and ground truth
- Dice Score: Measures similarity between predicted and actual segmentation
- Pixel Accuracy: Percentage of correctly classified pixels

Challenges and Solution

1. **Similar Appearance of Classes**
   - Classes like dry grass, bushes, and ground clutter looked similar.
   - Solution: Used data augmentation and a robust segmentation model.
2. **Class Imbalance**
   - Some classes (sky, landscape) appeared more often than others (flowers, logs).
   - Solution: Applied weighted loss and Dice loss to improve detection of smaller classes.
3. **Boundary Detection Issues**
   - Model struggled to classify pixels at object boundaries.
   - Solution: Used higher-resolution images and more training epochs.
4. **Overfitting**
   - Model performed better on training data than validation data.
   - Solution: Used data augmentation and validation monitoring.
5. **Computational Constraints**
   - Training required high computational resources.
   - Solution: Optimized image resolution and batch size.

Conclusion

This project developed a semantic segmentation model to identify and classify different elements in desert environments such as trees, bushes, rocks, dry grass, and sky. Using data preprocessing, augmentation, and model training, the system learned important visual patterns from the dataset. The model's performance was evaluated using the IoU metric, showing effective segmentation of desert scenes and accurate terrain classification.

Future scope

- Use more advanced segmentation models to improve accuracy and IoU score.
- Increase dataset size and diversity to improve model generalization.
- Apply advanced data augmentation techniques for better robustness.
- Improve boundary detection to reduce misclassification at object edges.
- Optimize the model for real-time deployment in autonomous vehicles.
- Integrate the system with robotics and off-road navigation platforms.